LABSHEET 6

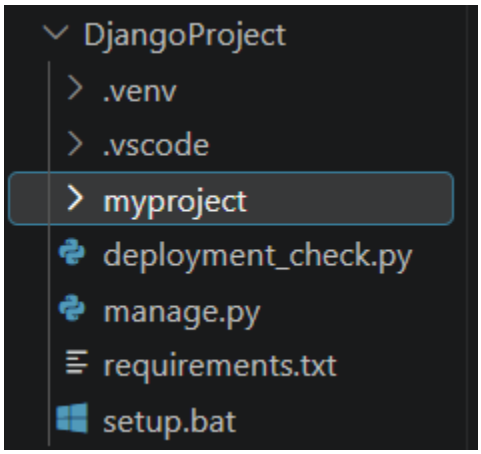
Demonstrate installation and workspace environment path mappings for Python runtimes, Django frameworks, and Visual Studio Code IDE setups.

Task 6.1: Setup an isolated Django virtual execution ecosystem using venv clusters and pipe project requirement definitions into a formal setup configuration script.

Task 6.2: Map custom user shortcuts and syntax automation workflows inside the VS Code development profile directory specifically for Python PEP8 compliance.

Task 6.3: Script an automated deployment check file that validates systemic runtime pathways, framework dependencies, and global server environment keys.

Project Directory



setup.bat

```
@echo off
python -m venv venv
call venv\Scripts\activate
python -m pip install --upgrade pip
pip install -r requirements.txt
echo Django environment setup completed.
```

settings.json configuration

```
{
    "python.analysis.typeCheckingMode": "basic",
    "editor.formatOnSave": true,
    "editor.rulers": [79],
```

```
    "editor.tabSize": 4,
    "editor.insertSpaces": true,
    "files.trimTrailingWhitespace": true,
    "editor.defaultFormatter": "ms-python.black-formatter"
}
```

deployment_check.py

```
import os
import sys
import importlib.util
def check_python():
    print("Python:", sys.version)
    print("Python Path:", sys.executable)
def check_django():
    if importlib.util.find_spec("django"):
        import django
        print("Django:", django.get_version())
    else:
        print("Django: NOT INSTALLED")
def check_environment():
    keys = ["DJANGO_SETTINGS_MODULE", "SECRET_KEY"]
    for key in keys:
        if os.environ.get(key):
            print(f"{key}: OK")
        else:
            print(f"{key}: MISSING")
print("=== Deployment Check ===")
check_python()
check_django()
check_environment()
```

```
=== Deployment Check ===
Python: 3.13.13 (tags/v3.13.13:01104ce, Apr  7 2026, 19:25:48) [MSC v.1944 64 bit (AMD64)]
Python Path: E:\full stack\DjangoProject\venv\Scripts\python.exe
Django: 6.1
DJANGO_SETTINGS_MODULE: OK
SECRET_KEY: OK
(venv) PS E:\full stack\DjangoProject>
```